

No announcements this week

1: Double integral evaluations

(a) $\iint_R \left(y + \frac{x}{y^2}\right), R = \{0 \leq x \leq 2, 1 \leq y \leq 2\}$

For rectangular regions, just place bounds directly & make sure order of dx & dy match

$$= \int_1^2 \int_0^2 \left(y + \frac{x}{y^2}\right) dx dy =$$

$$\int_1^2 \left(xy + \frac{x^2}{2y^2}\right) \Big|_{x=0}^{x=2} dy = \int_1^2 \left(2y + \frac{2}{y^2}\right) dy$$

$$= \left(y^2 - \frac{2}{y}\right) \Big|_1^2 = (4-1) - (1-2) = 4.$$

(b) This is $\int_{-2}^3 \int_0^{\pi/2} x \cos^2 y dy dx =$

$$\left(\int_{-2}^3 x dx\right) \left(\int_0^{\pi/2} \cos^2 y dy\right) := AB.$$

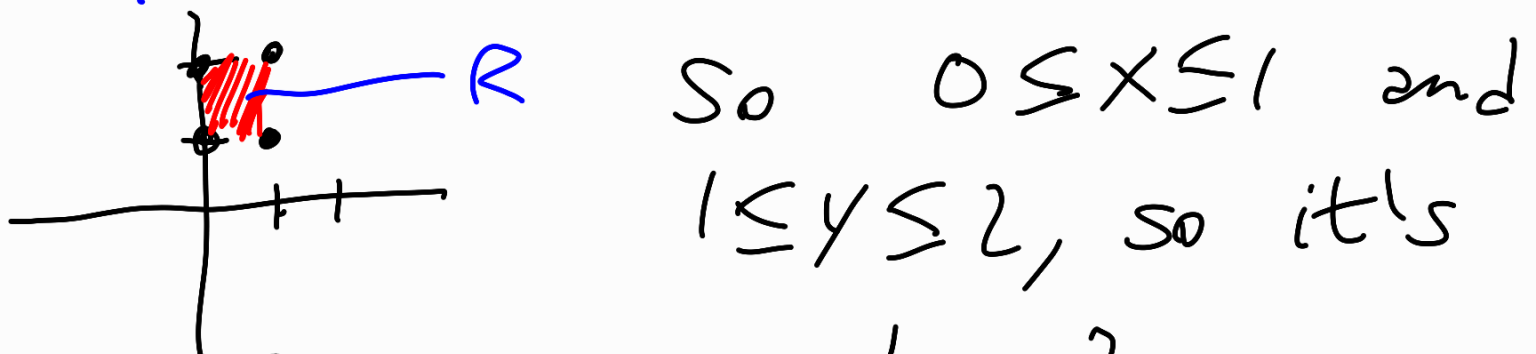
$$A = \frac{x^2}{2} \Big|_{-2}^3 = \frac{3^2 - (-2)^2}{2} = \frac{5}{2}$$

$$B = \int_0^{\pi/2} \frac{\cos(2y) + 1}{2} dy = \frac{1}{2} \left(\frac{1}{2} \sin(2y) + y \right) \Big|_0^{\pi/2}$$

$$= \frac{1}{2} \left(\frac{1}{2} \cdot 0 + \frac{\pi}{2} \right) - \frac{1}{2} \cdot 0 = \frac{\pi}{4}$$

so integral = $AB = \frac{5\pi}{8}$.

(c) First, plot the rectangle.



So $0 \leq x \leq 1$ and $1 \leq y \leq 2$, so it's

$$\int_0^1 \int_1^2 \frac{dy}{(2x+3y)^2} dx = \int_0^1 \frac{1}{9} \int_1^2 \frac{dy}{(y + \frac{2}{3}x)^2} dx$$

$$= \int_0^1 -\frac{1}{9} \frac{1}{(y + \frac{2}{3}x)} \Big|_{y=1}^{y=2} dx =$$

$$\int_0^1 \frac{1}{9} \left(\frac{1}{1 + \frac{2}{3}x} - \frac{1}{2 + \frac{2}{3}x} \right) dx =$$

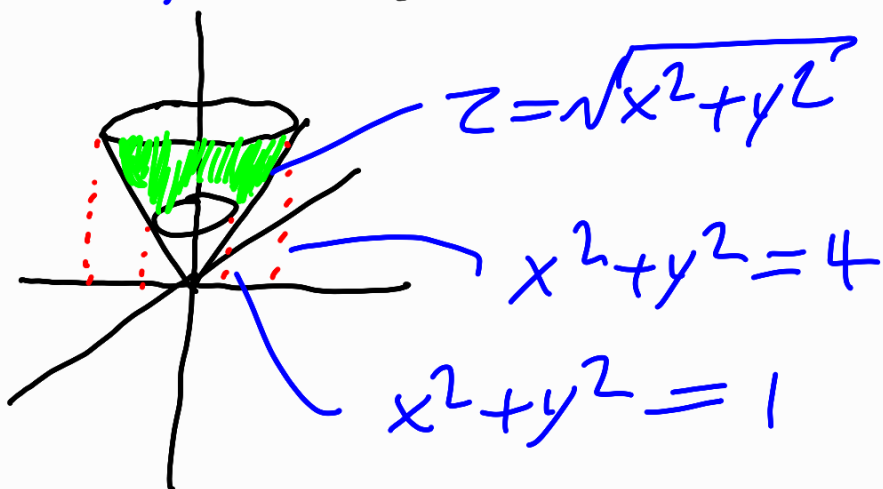
$$\frac{1}{9} \frac{3}{2} \int_0^1 \left(\frac{1}{\frac{3}{2}+x} - \frac{1}{3+x} \right) dx =$$

$$\frac{1}{6} \ln \left| \frac{\frac{3}{2}+x}{3+x} \right| \Big|_0^1 = \frac{1}{6} \left(\ln \frac{5/2}{4} - \ln \frac{1}{2} \right)$$

$$= \frac{1}{6} \ln \frac{5}{4}.$$

7: Volume under $z = \sqrt{x^2 + y^2}$
and above $1 \leq x^2 + y^2 \leq 4$.

Step 1: graph region, find bounds



Recall $x = r \cos \theta$, $y = r \sin \theta$ for polar,
so $x^2 + y^2 = r^2 \Rightarrow \underline{1 \leq r \leq 2}$. Since
we are under cylinder, $z \leq \sqrt{x^2 + y^2}$

= r . The region is rotationally symmetric, so no θ constraints, so $0 \leq \theta \leq 2\pi$.

Step 2: Plug in bounds & dV .

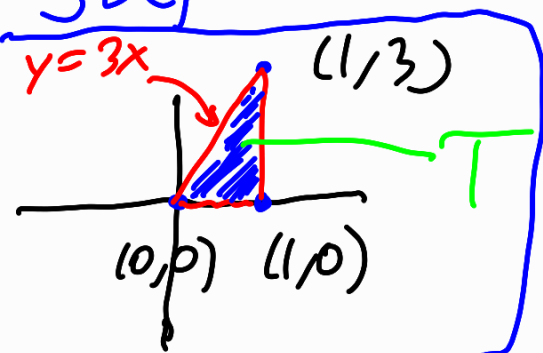
Recall $dV = r dr d\theta dz$, so

$$\text{Volume} = \int_0^2 \int_0^{2\pi} \int_0^r r dz dr d\theta =$$

$$2\pi \int_0^2 r^2 dr = \frac{2\pi}{3} r^3 \Big|_0^2 = \frac{16\pi}{3}.$$

5: Average $f=xy$ over triangle with vertices $(0,0), (1,0), (1,3)$.

Step 1: Graph & find bounds



Diagonal line of T is the line $y=3x$,

$$\text{so } \underline{0 \leq y \leq 3x} \text{ \& \underline{0 \leq x \leq 1}}$$

Step 2: Recall average formula

Recall: $\text{avg}_T f = \frac{1}{\text{area}(T)} \int_T f dA.$

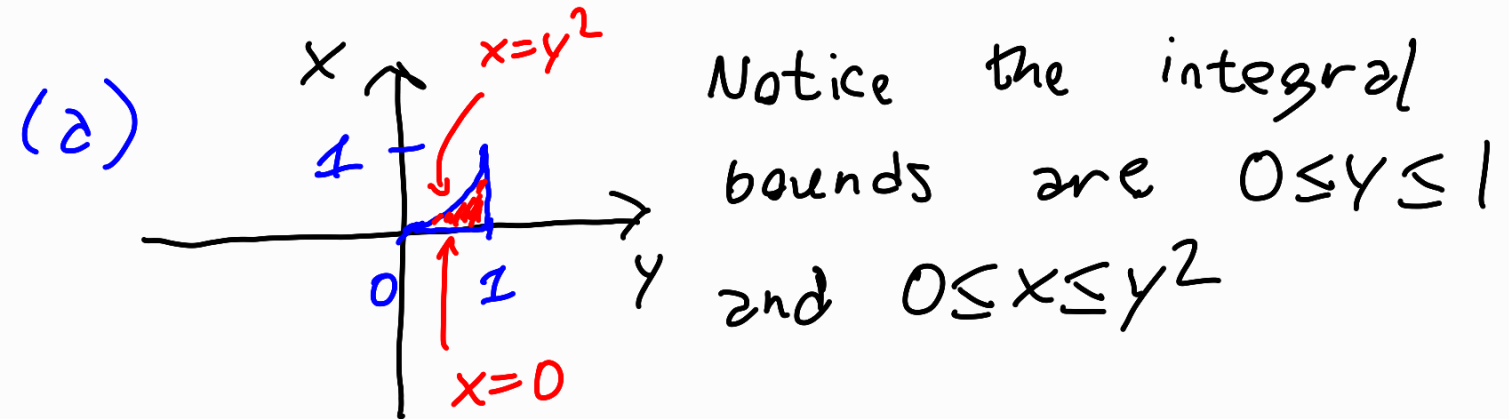
$$|T| = \frac{1}{2}bh = \frac{1}{2} \cdot 1 \cdot 3 = \frac{3}{2}.$$

$$\int_T f dA = \int_0^1 \int_0^{3x} xy \, dy \, dx = \int_0^1 x \frac{(3x)^2}{2} dx = \frac{9}{2} \int_0^1 x^3 dx = \frac{9}{8}.$$

$$\text{So average} = \frac{9/8}{3/2} = \frac{3}{4}.$$

Note! you can check your work for average value problems by seeing if your answer is close to an estimate. On T , $0 \leq x \leq 1$ and $0 \leq y \leq 3$, so we expect $x \approx \frac{1}{2}$, $y \approx \frac{3}{2}$ on average, so $f = xy \approx \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}.$

6: $\int_0^1 \int_0^{y^2} f(x,y) dx dy$: Sketch region,
reverse order, find for $f = \sqrt{x} + 3y$



(b) To reverse the bounds, we need to find constant bounds for x . From the graph, $0 \leq x \leq 1$. For y , notice that $x \leq y^2 \Rightarrow \sqrt{x} \leq y$. Also, $y \leq 1$ still. So new order is

$$\int_0^1 \int_{\sqrt{x}}^1 f(x,y) dy dx$$

(c) This is $\int_0^1 \int_{\sqrt{x}}^1 (\sqrt{x} + 3y) dy dx =$

$$\int_0^1 (y\sqrt{x} + 1.5y^2) \Big|_{\sqrt{x}}^1 = \int_0^1 (1 - \sqrt{x})\sqrt{x} + 1.5[1^2 - \sqrt{x}^2] dx$$

$$= \int_0^1 (\sqrt{x} - x + 1.5 - 1.5x) dx = \int_0^1 \sqrt{x} + 1.5 - 2.5x dx$$

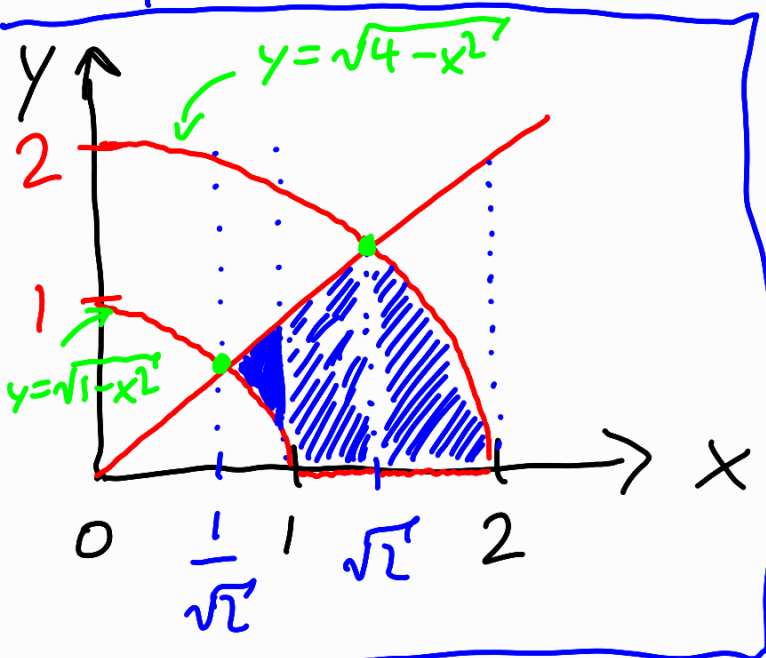
$$= \left(\frac{2}{3} x^{3/2} + 1.5x - 1.25x^2 \right) \Big|_0^1 = \frac{2}{3} + 1.5 - 1.25$$

$$= \frac{2}{3} + 0.25 = \frac{8}{12} + \frac{3}{12} = \frac{11}{12}$$

9: $\left(\int_{1/\sqrt{2}}^1 \int_{\sqrt{1-x^2}}^x + \int_1^{\sqrt{2}} \int_0^x + \int_{\sqrt{2}}^2 \int_0^{2\sqrt{4-x^2}} \right) xy dy dx$

combine into one integral & evaluate

Step 1: Graph 3 regions to find bounds



For the 1st integral,

$$\frac{1}{\sqrt{2}} \leq x \leq 1 \quad \text{and}$$

$$\sqrt{1-x^2} \leq y \leq x, \quad \text{so}$$

we will draw the curves $\frac{1}{\sqrt{2}} = x$, $1 = x$,

$y = \sqrt{1-x^2}$, $y = x$. Similarly, the other 2 double integrals are 4 more curves.

After graphing, we see that the 3 regions combine to a circular sector (in blue) with inner, outer radius 1, 2, so

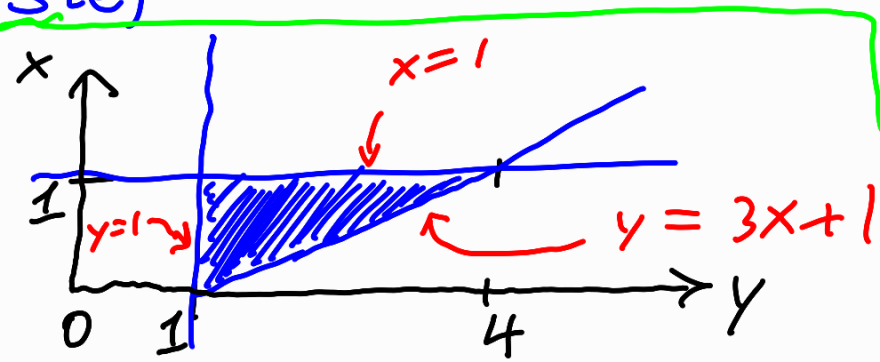
$1 \leq r \leq 2$. The 2 lines bounding the region are $y=0$, $y=x$, corresponding to $\theta=0^\circ, 45^\circ$, so $0 \leq \theta \leq \frac{\pi}{4}$

Step 2: Substitute bounds & evaluate

$$\begin{aligned}
 xy \, dy \, dx &= (r \cos \theta)(r \sin \theta) r \, dr \, d\theta = \\
 r^3 \sin \theta \cos \theta \, dr \, d\theta &= \frac{1}{2} r^3 \sin 2\theta \, dr \, d\theta, \text{ so} \\
 \text{integral is } \int_0^{\pi/4} \int_1^2 \frac{1}{2} r^3 \sin 2\theta \, dr \, d\theta &= \\
 \left(\int_0^{\pi/4} \sin 2\theta \, d\theta \right) \left(\int_1^2 0.5 r^3 \, dr \right) &= \left(-\frac{1}{2} \cos 2\theta \Big|_0^{\pi/4} \right) \left(\frac{r^4}{8} \Big|_1^2 \right) \\
 &= -\frac{1}{2} \left(\cos \frac{\pi}{2} - \cos 0 \right) \left(\frac{15}{8} \right) = \frac{15}{16}.
 \end{aligned}$$

3: R bounded by $y=3x+1$, $x=1$, $y=1$. Draw R & setup $\iint_R f \, dA$ as type I & II.

Step 1: Graph R & find bounds



From the graph,
 $0 \leq x \leq 1$ and
 $1 \leq y \leq 3x+1$.

Also, we can use constant bounds on y instead. Then $1 \leq y \leq 4$, but $x \leq 1$ still.
 $y \leq 3x+1 \Rightarrow \frac{y-1}{3} \leq x$, so $\frac{y-1}{3} \leq x \leq 1$

Step 2: Now just substitute bounds.

$$\iint_R f dA = \int_0^1 \int_1^{3x+1} f dx dy \text{ or } \int_1^4 \int_{\frac{y-1}{3}}^1 f dx dy$$

are the Type I/II integrals.
